

MATHEMATICKS

The Expert holds this manual. The Defuser holds the device.

The Defuser may not read this manual. You may not look at the device. Modules are not named on the device — find yours by what the Defuser describes to you.

BEFORE YOU START

1. Agree who is the Defuser and who is the Expert.
2. The Expert reads a module's section aloud; the Defuser reads the device aloud.
3. Three mistakes and the device goes off. So does the clock running out.

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Triangles

HARD

What's math without trigonometry?

A triangle is displayed on the panel, with a spec plate beneath it carrying three lengths. Below the display are three buttons: **red**, **green**, and **blue**. To the right of the display are four directional buttons, arranged like a controller: **up**, **down**, **left**, and **right**.

All buttons must be pressed in the **correct order** to disarm this module. Pressing the **wrong button**, or pressing the correct buttons **out of order**, will strike you.

This is a **hard** module.

Complete **Step 1** in full before moving on to **Step 2**.

Step 1: The Colored Buttons

Note the triangle's classification **by its sides** (equilateral, isosceles, or scalene), and the current left-to-right order of the colored buttons. Follow the rules **in the order they are listed** and perform the **first action that applies**.

- If the triangle is **equilateral**, press the button in the 2nd position, then the 1st, then the 3rd.
- If the triangle is **isosceles**, press the button in the 3rd position, then the 2nd, then the 1st.
- If the triangle is **scalene**, press each button in the order they appear, left to right.

Now read the three lengths on the spec plate beneath the triangle. Add the two shorter lengths together. If their sum is GREATER than the longest length, the three lengths can form a triangle: press the three buttons in the order you just worked out. If their sum is equal to or less than the longest length, they cannot form a triangle: press those same three buttons in REVERSE order.

Step 2: The Directional Buttons

Note the triangle's three angle measures. Subtract the smallest angle from the largest angle. This value is called **D**.

Now determine **X** using the triangle's classification **by its angles**, following the first rule that applies:

- If the triangle is **right**, X is 10.
- If the triangle is **acute**, X is 20.
- If the triangle is **obtuse**, X is 30.

Now modify X using D, following the first rule that applies:

- If D is 0, leave X the same.
- If D is between 1 and 30, add D to X.
- If D is between 31 and 60, double X.
- If D is greater than 60, subtract D from X.

This final value is called **Z**, and you will use it soon.

Attention!

Take note of the **original left-to-right order** the colored buttons were in **before** you pressed them in Step 1. If that order was **not** red-green-blue, add 5 to Z before continuing.

Now follow the rules below to see which order to press the **directional** buttons. The numbers in the diagrams refer to the **order you press** each direction.

- If Z is 20 or below, refer to **Diagram 1**.
- If Z is from 21 to 40, refer to **Diagram 2**.
- If Z is from 41 to 60, refer to **Diagram 3**.
- If Z is 61 or above, refer to **Diagram 4**.

	Diagram 1	Diagram 2	Diagram 3	Diagram 4
Up	1	2	3	4
Right	2	3	4	1
Down	3	4	1	2
Left	4	1	2	3

Venn Diagram

EASY

Every number belongs somewhere. So do you.

The panel shows three overlapping circles, **A**, **B** and **C**, inside a box marked **U**. Numbers are scattered across the eight regions, and **every number is a button**. There is also a separate **empty set button**, marked $\emptyset$.

Below the circles is a **set expression**. Your job is to work out which numbers that expression describes, then press them.

Pressing a **wrong number**, or pressing the correct numbers **out of order**, will strike you.

This is an easy module.

Step 1: Reading the Panel

The eight regions are numbered as shown below. The defuser reads out the numbers **one region at a time**, and the expert writes them down. The defuser must also read out the set expression exactly as it appears.

Region	Numbers that are in	Region	Numbers that are in
1	A only	5	A and C, but not B
2	B only	6	B and C, but not A
3	C only	7	all three
4	A and B, but not C	8	none of the three

Step 2: Reading the Symbols

Symbol	Name	Meaning
U	Union	In one set, the other, or both
$\cap$	Intersection	Only what is in both sets
'	Complement	Everything not in that set, region 8 included
()	Brackets	Work out what is inside them first

Work out **brackets** first, then **complements**, then **intersections**, and **unions** last.

Step 3: Finding the Target Set

Go through the regions **one at a time, from 1 to 8**. For each region, ask whether the expression describes it. If it does, every number in that region belongs to your **target set**. If it does not, ignore that region completely.

For example, $A \cap B'$ describes every number that is in A but not in B. That is region 1 and region 5, so the target set is every number sitting in those two regions.

Step 4: Pressing the Buttons

Press every number in the target set in **ascending order**. Press nothing else.

If the target set turns out to be **empty**, press the **empty set button** once and nothing else.

Rationality

MEDIUM

Be real. Be rational.

A number is displayed on the panel. Below the display are four buttons arranged in a **diamond**. Each button is a different colour: **yellow**, **blue**, **green**, and **red**. The defuser must read them out starting from the **top** button and going **clockwise**.

This module takes **five presses**. A button may be pressed more than once, and some buttons may never be pressed at all.

All buttons must be pressed in the **correct order** to disarm this module. Pressing the **wrong button**, or pressing the correct buttons **out of order**, will strike you.

This is a **medium** module.

Complete **Step 1** in full before moving on to **Step 2**. **Do not press anything until Step 2 is finished.**

Step 1: Reading the Number

Answer the five questions below **in the order they are listed**. Each answer gives you one colour. Write the five colours down from left to right, this is the **base sequence**.

	Question	If yes	If no
1	Is the number real?	Red	Blue
2	Is the number rational?	Yellow	Green
3	Is the number an integer?	Green	Red
4	Is the number negative?	Blue	Yellow
5	Is the number a natural (counting) number?	Red	Blue

Take note: **0** is a whole number, but it is **not** a natural number.

Step 2: Turning the Diamond

Note the **serial number**. Follow the rules **in the order they are listed** and perform the **first action that applies**.

- If the serial contains a digit **7 or higher** and a digit **3 or lower**, replace every colour in the base sequence with the colour **opposite** it in the diamond.
- Otherwise, if the serial contains a digit **7 or higher**, replace every colour with the colour **one position clockwise**.
- Otherwise, if the serial contains a digit **3 or lower**, replace every colour with the colour **one position counter-clockwise**.
- If none of the above apply, leave the base sequence **exactly as it is**.

Opposite means top to bottom and left to right. Clockwise and counter-clockwise always follow the diamond **as it is arranged for the round**, not the colours themselves.

Press the buttons in the final five-colour order you worked out.

Angles

HARD

Everything comes down to timing.

Two angles are drawn together on the panel so that their relationship is visible, with both measures labelled. Below them is a **colour panel** that changes colour every **two seconds**, and a single **button**.

The button must be pressed **while the correct colour is showing**. Pressing on the wrong colour will strike you. Letting the correct colour pass costs you time, but it is not a strike.

This is a **hard** module.

Step 1: The Two Angles

Classify each angle and read off its digit. Write the **first angle's digit**, then the **second angle's digit**. Together they are **X**.

Angle type	Measure	Digit
Zero	exactly 0 degrees	6
Acute	more than 0, less than 90	4
Right	exactly 90 degrees	1
Obtuse	more than 90, less than 180	0
Straight	exactly 180 degrees	5
Reflex	more than 180, less than 360	2
Complete	exactly 360 degrees	3

Step 2: The Angle Pair

Follow the rules **in the order they are listed** and take the **first one that applies**. This gives you **Y**.

- If the two angles are drawn as **vertical angles**, formed by two crossing lines, Y is **3**.
- Otherwise, if the two measures add up to exactly **90 degrees**, they are **complementary** and Y is **2**.
- Otherwise, if they add up to exactly **180 degrees**, they are **supplementary** and Y is **1**.
- If none of the above apply, Y is **4**.

Step 3: The Color

Write the **two digits of X**, then the **digit of Y** after them. Read all three as one number. That is **Z**. Leading zeros count for nothing, so 0, 4, 2 is simply **42**.

Z is from	Press on	Z is from	Press on
0 to 110	Red	331 to 440	Green
111 to 220	Orange	441 to 550	Blue
221 to 330	Yellow	551 to 664	Violet

Attention!

If **either** of the two angles is **reflex**, press on the **next colour down the list** instead. Violet wraps back around to red.

Sequences

EASY

The pattern continues with or without you.

A sequence of five terms is displayed on the panel, with one term left **blank**. Below the display are two input boxes, one for the **missing term** and one for the **sum**, and a **submit** button.

Every term is written in **letters**, not digits. Each letter stands for a single digit, and letters are read left to right, so a two-letter term is a two-digit number.

Both boxes must be filled in **correctly** before submitting. Submitting with **either box wrong** will strike you.

This is an **easy** module.

Answer in **digits**, never in letters.

Step 1: Decoding the Letters

Take note of the **serial number**. If it contains a **vowel**, read the letters using the bottom row of the table. If it does not, use the top row.

Letter	A	B	C	D	E	F	G	H	I	J
No vowel	0	1	2	3	4	5	6	7	8	9
Vowel	1	2	3	4	5	6	7	8	9	0

For example, with **no vowel** in the serial, BB is 11 and CA is 20. With a **vowel** in the serial, those same terms become 22 and 31.

Step 2: The Missing Term

Decode the four terms you were given and find the pattern. Follow the rules **in the order they are listed** and use the **first one that applies**.

- If the **difference** between each pair of consecutive terms is the same, the sequence is **arithmetic**. Keep adding that difference.
- If the **ratio** between each pair of consecutive terms is the same, the sequence is **geometric**. Keep multiplying by that ratio.

The blank is not always the last term. Work out the pattern from the terms you can see, then fill the gap. Enter this value in the **first** box.

Step 3: The Sum

Add together **all five terms**, including the one you just worked out. Enter this value in the **second** box.

Attention!

If the sum is **100 or greater**, enter only its **last two digits**. A sum of 143 is entered as 43, and a sum of 100 is entered as 00.

Once both boxes are filled in, press **submit**.

Unit Conversions

HARD

Measure twice. You only get one press.

The module has a **top panel** and a **bottom panel**. To their left are **three squares** that light up, and below those an **answer box** with a **submit** button.

One panel shows an **everyday object**. The other shows that object sitting on a **measuring instrument**: a ruler, a weighing scale, or a graduated cylinder. Sometimes both appear in the same panel and the other panel is **dark**.

The instrument is marked with **numbers only, no units**. The defuser reads off the number. The lights decide what that number means.

Type the converted value into the answer box and press **submit**. A wrong answer will strike you. The answer comes out **exactly** (a whole number), so if you are left with a long trail of decimals, something has gone wrong.

This is a **hard** module.

Step 1: What Is Being Measured

The instrument tells you the category. A **ruler** means length, a **weighing scale** means mass, and a **graduated cylinder** means volume.

Step 2: Which Direction

The layout of the two panels tells you which way the conversion runs.

Layout	Top panel	Bottom panel	Conversion
A	Object	Instrument	Metric to American
B	Instrument	Object	American to metric
C	Object and instrument	Dark	American to American
D	Dark	Object and instrument	Metric to metric

Step 3: Which Units

Every category has a **small**, a **middle** and a **large** unit on each side. The lit squares tell you which two are involved.

Squares lit	Layouts A and B	Layouts C and D
1st only	Middle to middle	Middle to large
2nd only	Middle to large	Large to middle
3rd only	Middle to small	Middle to small
1st and 2nd	Small to middle	Small to middle
2nd and 3rd	Small to large	Large to small
All three	Small to small	Small to large

In layout **A** read the pair as written, from metric to American. In layout **B** read the same pair **backwards**, from American to metric. In **C** and **D** read the pair as written.

The 1st and 3rd squares are never lit together, and the squares are never all dark.

Step 4: The Unit Charts

The number beside each unit is its size in that category's **base unit**, which is the centimetre for length, the gram for mass, and the millilitre for volume.

Metric	Small	Middle	Large
Length	centimetre (1)	metre (100)	kilometre (100 000)
Mass	milligram (0.001)	gram (1)	kilogram (1 000)
Volume	millilitre (1)	litre (1 000)	kilolitre (1 000 000)
American	Small	Middle	Large
Length	inch (2.54)	foot (30.48)	yard (91.44)
Mass	ounce (28.35)	pound (453.6)	ton (907 200)
Volume	cup (236.6)	quart (946.4)	gallon (3785.6)

Step 5: Converting

Multiply the reading by the **source unit's number**, then divide by the **target unit's number**.

For example, a reading of 152.4 with a source of centimetre (1) and a target of foot (30.48) gives 152.4 times 1, divided by 30.48, which is exactly **5**.

Attention!

Instruments are read carefully or not at all.

- Measure from the ruler's **zero mark**, not from the end of the ruler.
- Read the weighing scale where the **centre of the pointer** falls, not its edge.
- Read the cylinder at the **bottom of the curve** of the liquid, at eye level.
- If the reading sits **between two marks**, take the **lower** mark.

Parallel Lines Cut by a Transversal

MEDIUM

Parallel lines never meet. Neither do second chances.

Two parallel lines cut by a transversal are shown on the panel, forming **eight angles**. Each angle sits on a **button**, numbered 1 to 8 as shown below. Exactly one angle has its measure displayed.

Four of the eight buttons must be pressed in the **correct order** to disarm this module. Pressing the **wrong button**, or pressing the correct buttons **out of order**, will strike you.

This is a **medium** module.

At each intersection the angles are numbered **clockwise, starting from the upper-left**. The angle positions never change, only the measure that is displayed.

Step 1: The Two Measures

Because the lines are parallel, only **two** different measures appear across all eight angles. Angles **1, 3, 5 and 7** always share one measure. Angles **2, 4, 6 and 8** always share the other, and the two measures add up to **180 degrees**.

Work out **X**, the measure of **angle 1**. If the displayed angle is odd-numbered, X is simply the measure shown. If it is even-numbered, X is **180 minus** the measure shown.

Step 2: The Starting Angle

Let **N** be the number of the angle whose measure was displayed. Follow the rules **in the order they are listed** and take the **first one that applies**. This gives you the starting angle, **S**.

- If X is exactly **90**, S is angle **2**.
- If X is **acute** and N is **odd**, S is angle **1**.
- If X is **acute** and N is **even**, S is angle **4**.
- If X is **obtuse** and N is **odd**, S is angle **6**.
- If X is **obtuse** and N is **even**, S is angle **7**.

Step 3: The Press Order

Find your starting angle S in the table below. Press **S first**, then its **corresponding** angle, then its **vertical** angle, and last its **same-side** angle.

S	Corresponding	Vertical	Same-side
1	5	3	8
2	6	4	7
4	8	2	5
6	2	8	3
7	3	5	2

Corresponding and vertical angles are **equal** to S. The same-side angle is **supplementary** to S, so the two of them add up to 180 degrees.

Attention!

Take note of the **serial number**. If its **last character is an even digit**, press the four buttons in **reverse order** instead. If the last character is a letter, this rule does not apply.

Mutually Exclusive Events

MEDIUM

Some things cannot happen together.

Two playing cards are displayed on the panel, drawn from **two separate decks**, so the two cards may be identical. Below the display is a text box and a **submit** button.

The defuser only needs to read out the **suit** and **rank** of each card. The expert works out everything else.

Card values are: **Ace is 1, Jack is 11, Queen is 12, King is 13**, and every other card is worth its face value.

You will build a **four-digit binary string**, convert it to decimal, and type that number into the box. A **wrong answer** will strike you.

This is a **medium** module.

Step 1: Building the Binary String

Two events are **mutually exclusive** when they cannot both happen at the same time. Imagine a single card being drawn. For each pair of events below, ask whether one card could satisfy **both** events at once.

Write **1** if the pair is mutually exclusive. Write **0** if it is not. Write the digits down from left to right.

Digit	First event	Second event
1	The card is Card 1's suit	The card is Card 2's suit
2	The card is Card 1's colour	The card is Card 2's colour
3	The card is Card 1's rank	The card is Card 2's rank

The **fourth digit** is not a mutual exclusivity test. Add the two card values together. Write **1** if the sum is **13 or more**, and **0** if it is less than 13.

Step 2: Converting to Decimal

Each position in the binary string has a **place value**. Add up the place values of every position holding a **1**, and ignore every position holding a **0**.

Position	1st	2nd	3rd	4th
Place value	8	4	2	1

For example, **1011** is 8 plus 2 plus 1, which is **11**. Your answer will always be a whole number from **0 to 15**.

Attention!

Take note of the **serial number**. If its **last character is an odd digit**, reverse the order of the four binary digits **before** converting. A string of 1011 becomes 1101, which is **13**, not 11. If the last character is a letter, this rule does not apply.

Type the decimal number into the box and press **submit**.